

Titanic Dataset

Exploratory Data Analysis

Findings Report

1. Objective

The objective of this analysis is to perform a thorough Exploratory Data Analysis (EDA) on the Titanic passenger dataset. The goal is to extract meaningful insights through visual and statistical exploration — understanding the distribution of features, identifying missing data, and uncovering relationships between variables, particularly those that influenced passenger survival.

2. Dataset Overview

The dataset contains 891 passenger records with 12 columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked. The target variable is Survived, where 0 indicates the passenger did not survive and 1 indicates survival.

The numerical features include Age, Fare, SibSp (siblings/spouses aboard), and Parch (parents/children aboard). Categorical features include Sex, Pclass, Embarked, and Cabin. The mean age of passengers was approximately 29.7 years, and the average fare paid was around 32 units. The youngest passenger was less than 1 year old, while the oldest was 80.

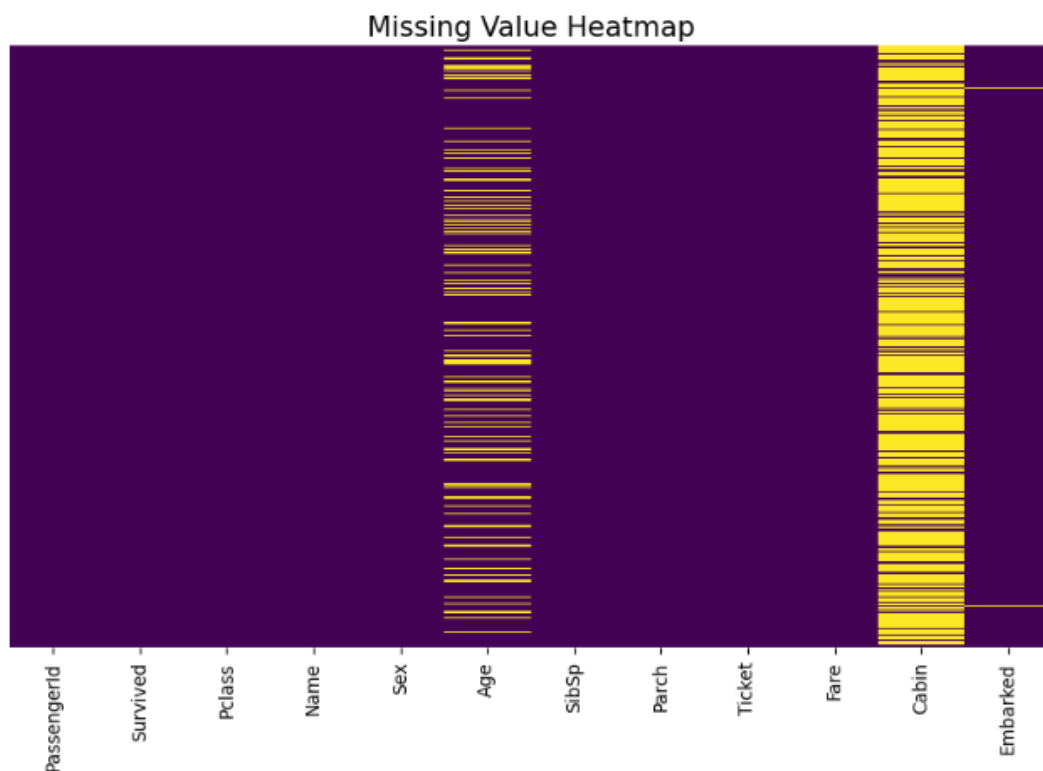
PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

3. Missing Value Analysis and Treatment

Three columns contained missing values. The Age column had 177 missing entries . The Cabin column had 687 missing entries , making it largely unusable. The Embarked column had only 2 missing entries .

Age was filled with the median value of 28 years, as the median is resistant to outliers. The Cabin column was dropped entirely due to its excessive missing rate. Embarked was filled with the mode value 'S' (Southampton), the most frequent port.



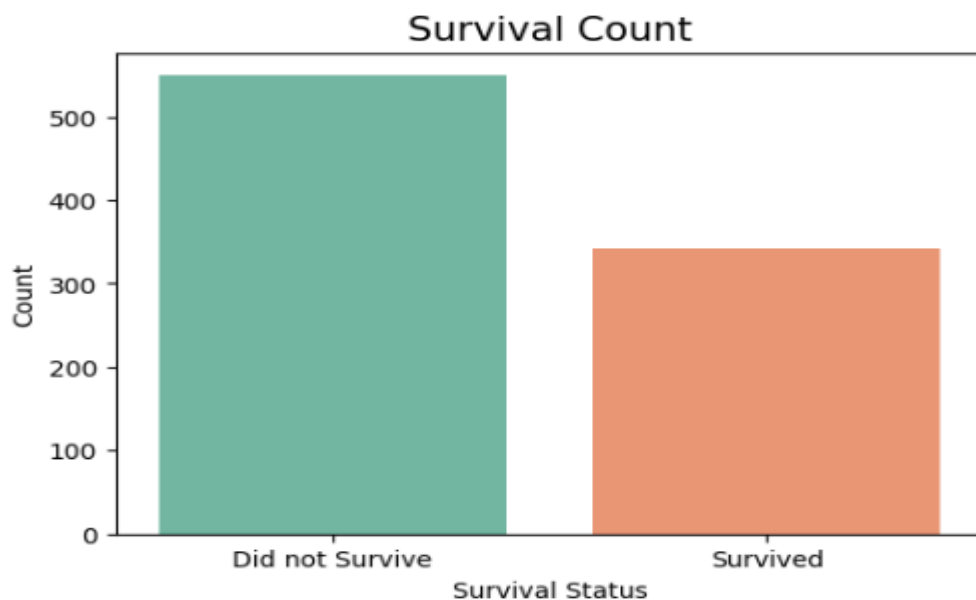
```
Missing values after cleaning:
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
dtype: int64
```

4. Univariate Analysis

Univariate analysis examines each variable individually to understand its distribution and frequency before exploring relationships between variables.

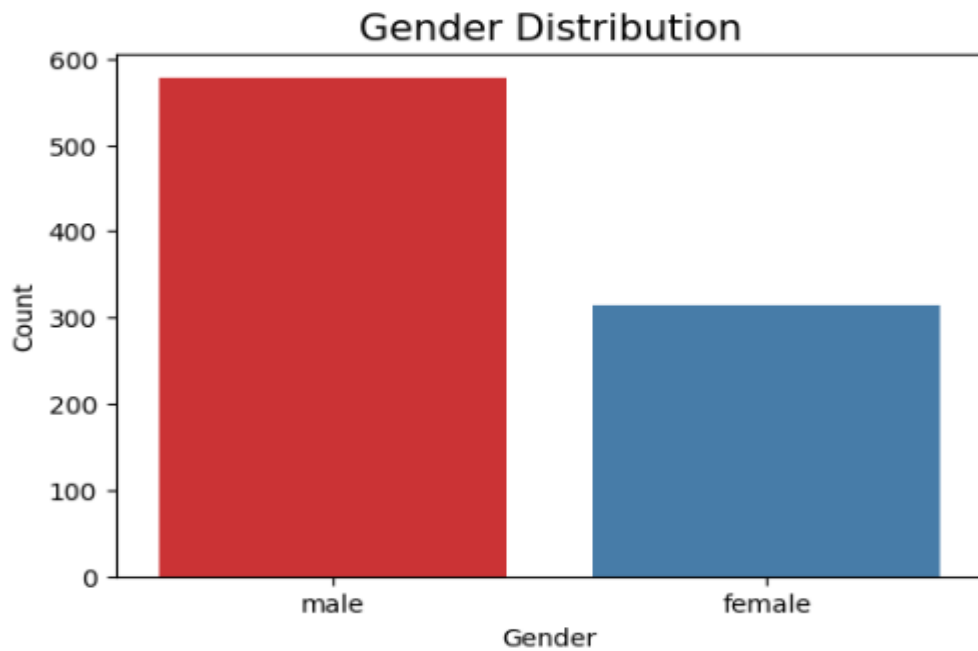
4.1 Survival Count

Out of 891 passengers, 549 did not survive and 342 survived . The majority of passengers perished, indicating a significant class imbalance to be considered in any future predictive modelling.



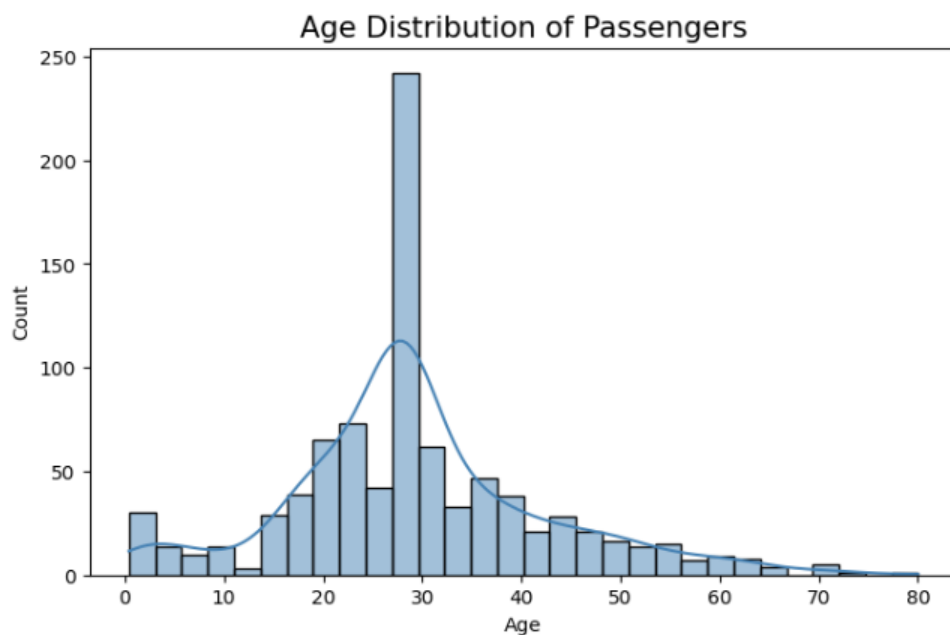
4.2 Gender Distribution

The dataset contains 577 male passengers and 314 female passengers . Males outnumbered females by nearly 2:1, which becomes especially significant when analysed alongside survival outcomes.



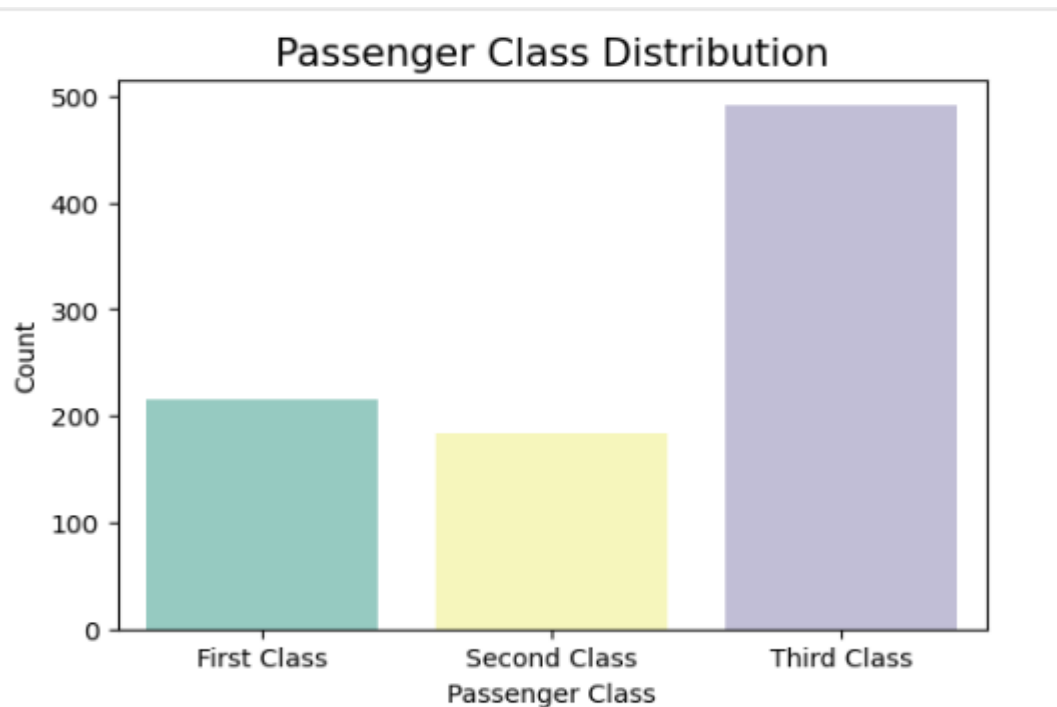
4.3 Age Distribution

The age distribution is slightly right-skewed, with most passengers aged between 20 and 35. The KDE curve peaks around 28 to 30 years. A small cluster of infants under 5 is visible, and passengers above 60 form a small minority.



4.4 Passenger Class Distribution

Third-class passengers were the largest group with 491 passengers, followed by first class with 216 and second class with 184. The majority of people aboard were travelling in the lowest class.

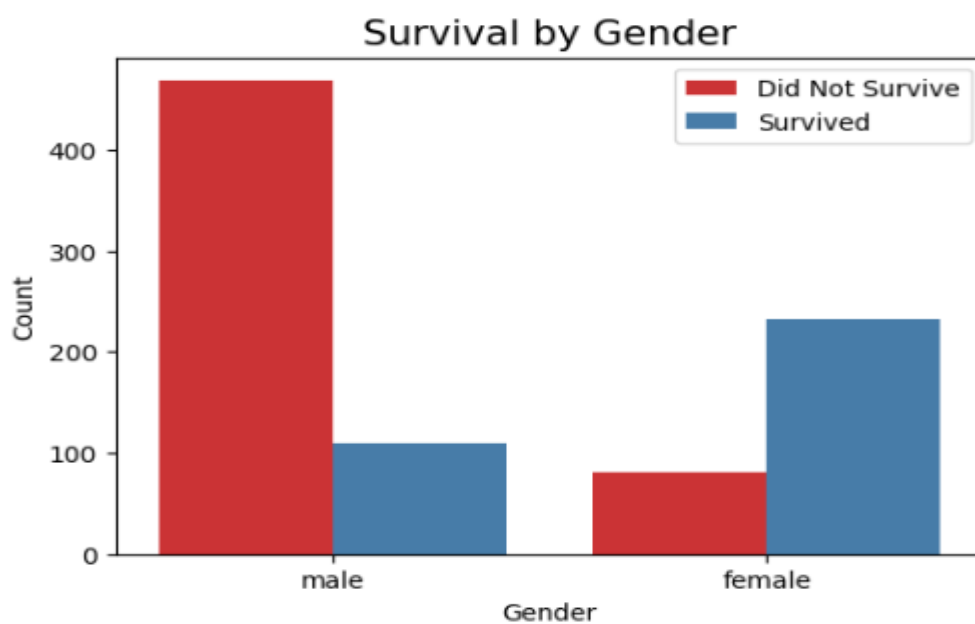


5. Bivariate Analysis

Bivariate analysis explores the relationship between two variables at a time, particularly how each feature relates to the survival outcome.

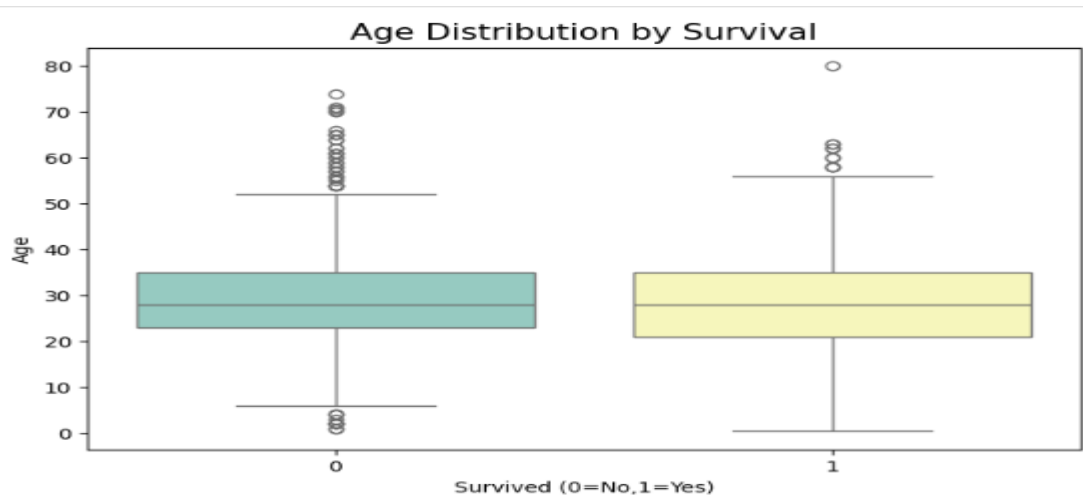
5.1 Survival by Gender

Gender was the most decisive factor in survival. Female passengers survived at approximately 235, while males survived at only around 110. This pattern strongly reflects the 'Women and Children First' evacuation protocol. The grouped bar chart clearly shows that most male passengers did not survive, while most female passengers did.



5.2 Age Distribution by Survival

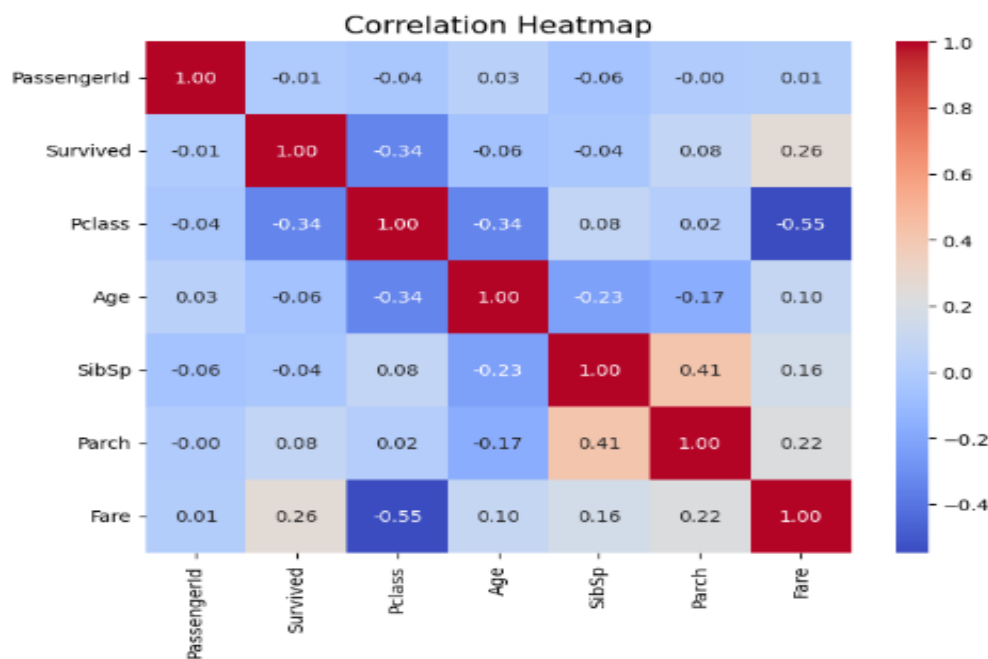
The boxplot comparing age between survivors and non-survivors shows both groups had a similar median age — approximately 28 for non-survivors and 30 for survivors. Age alone was not a strong differentiator. However, very young children appear slightly more in the survived group, consistent with evacuation priority given to children.



6. Multivariate Analysis

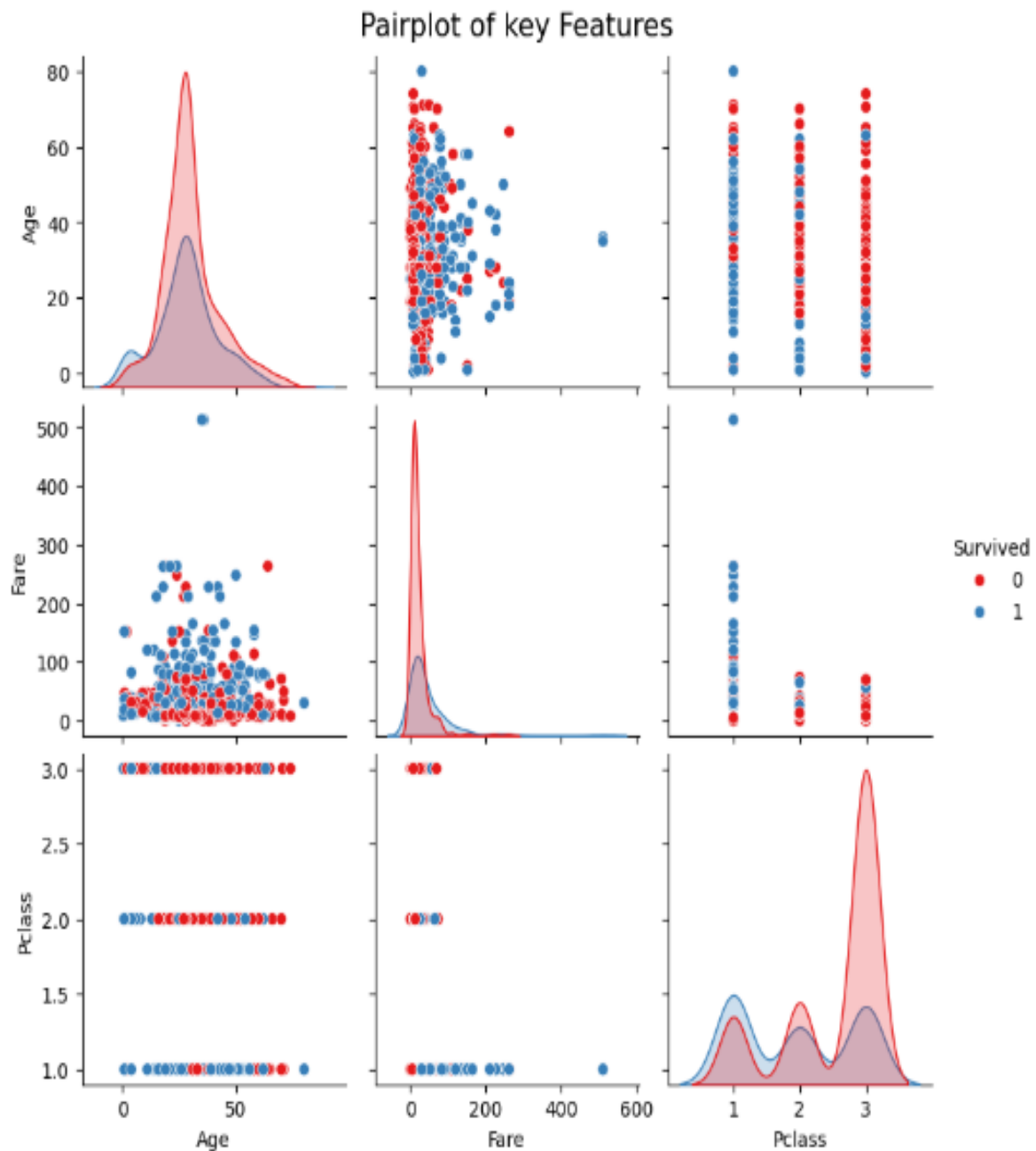
6.1 Correlation Heatmap

The correlation heatmap revealed several notable relationships. Pclass and Fare had the strongest correlation at -0.55, meaning higher-class passengers paid significantly more. Fare and Survived showed a positive correlation of 0.26 — passengers who paid more had better survival chances. Pclass and Survived had a correlation of -0.34, confirming that higher class was linked to better survival. SibSp and Parch had a moderate correlation of 0.41, reflecting that passengers travelling with family tended to have both.



6.2 Pairplot

The pairplot of Age, Fare, Pclass, and Survived coloured by survival status provided a simultaneous view of all pairwise relationships. Survivors clustered at lower Pclass values and higher Fare values. Age distributions overlapped significantly between the two groups, reinforcing that age was not a standalone predictor.



7. Key Findings

Gender was the strongest predictor of survival. Women survived at nearly four times the rate of men, directly reflecting the evacuation protocol followed during the disaster.

Passenger class was the second most important factor. First-class passengers had significantly higher survival rates than second or third class, pointing to the role of socioeconomic status in accessing lifeboats.

Fare paid was strongly correlated with passenger class and also showed a positive relationship with survival, reinforcing the socioeconomic dimension of the disaster.

Age was not a strong independent predictor of survival. Survivors and non-survivors had nearly identical age distributions.

The Cabin column was too sparse with over 77% missing values and was excluded. Age missing values were handled through median imputation to preserve the distribution.

8. Conclusion

This Exploratory Data Analysis of the Titanic dataset successfully identified the key factors that influenced passenger survival through systematic data profiling, missing value treatment, and univariate, bivariate, and multivariate exploration.

The three most important factors were gender, passenger class, and fare paid. Gender was the strongest predictor, with female passengers surviving at nearly four times the rate of males. Passenger class reflected socioeconomic differences that directly impacted lifeboat access. Age, while not a strong standalone predictor, may contribute in combination with other features in a predictive model.

The insights from this EDA provide a solid foundation for the next steps, including feature engineering and building classification models to predict survival outcomes.